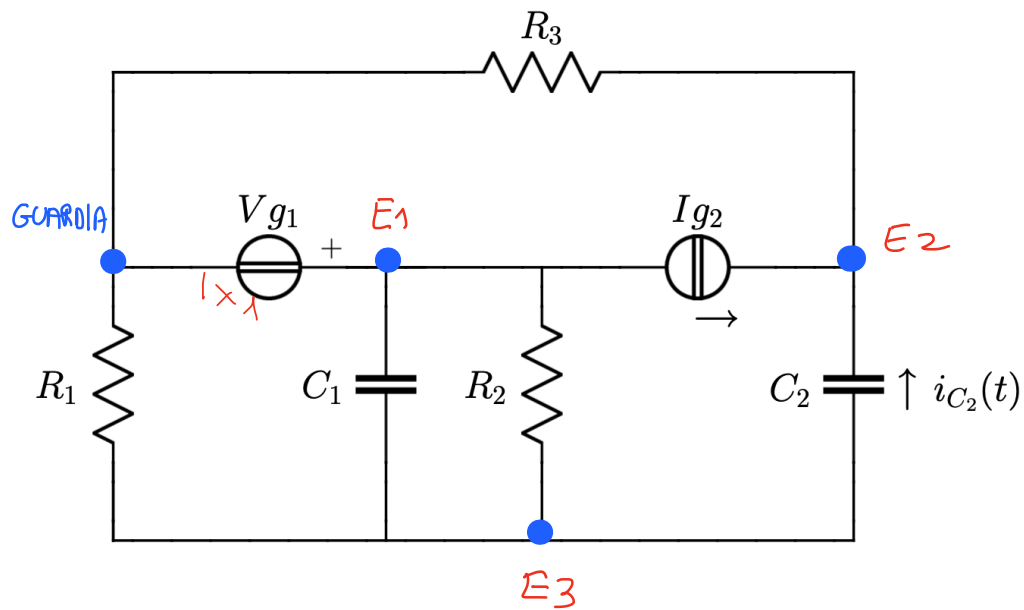


Parallelo NODI

domenica 3 agosto 2025 16:30



$$\begin{aligned} R_1 &= \frac{1}{2} \\ R_2 &= \frac{1}{2} \\ C_1 &= 4 \\ C_2 &= 2 \\ \mathbf{V}_{g1} &= -2 \\ \mathbf{I}_{g2} &= 1 + 8j \\ R_3 &= 1 \\ \omega &= 2 \end{aligned}$$

RISOLVO IL PARALLELO:

$$Y_a = j\omega C_1 + \frac{1}{R_2} = 8j + \frac{1}{\frac{1}{2}} = 8j + 2$$

$$Z_a = \frac{1}{Y_a} = \frac{1}{8j + 2} \cdot \frac{2 - 8j}{2 - 8j} = \frac{2 - 8j}{68} = \frac{1}{34} - \frac{2}{17}j$$

$$\begin{bmatrix} Y_a & 0 & -Y_a \\ 0 & \frac{1}{R_3} + j\omega C_2 & -j\omega C_2 \\ -Y_a & -j\omega C_2 & \frac{1}{R_1} + Z_a + j\omega C_2 \end{bmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix} = \begin{pmatrix} 1 \times 1 - I_{g2} \\ I_{g2} \\ 0 \end{pmatrix}$$

$$\begin{cases} y_{0a}E_1 - y_{0a}E_3 = I_{g1} - I_{g2} \\ \left(\frac{1}{R_3} + j\omega C_2\right)E_2 + (-j\omega C_2)E_3 = I_{g2} \\ -y_{0a}E_1 + (-j\omega C_2)E_2 + \left(\frac{1}{R_1} + y_{0a} + j\omega C_2\right)E_3 = 0 \\ E_1 = V_{g1} \rightarrow E_1 = -2 \end{cases}$$

RISOLVIAMO:

$$\begin{cases} -16j - 4 + (-8j - 2)E_3 = 1 \times 1 - 8j - 1 & 1 \\ (1 + 4j)E_2 + (-4j)E_3 = 1 + 8j & 2 \\ 16j + 4 + (-4j)E_2 + (2 + 2 + 8j + 4j)E_3 = 0 & 3 \end{cases}$$

DALLA 2 RITIRO E₃

$$-4jE_3 = 1 + 8j + (-1 - 4j)E_2$$

$$E_3 = \frac{-1}{4j} - \frac{8j}{4j} + \left(\frac{-1 - 4j}{-4j}\right)E_2$$

$$E_3 = \frac{1}{4}j - 2 + \left(-\frac{1}{4}j + 1\right)E_2$$

$$E_3 = -1$$

$$-16j - 4 + 8j + 2 = 1 \times 1 - 8j - 1$$

$$-16j - 4 + 8j + 2 + 8j + 1 = 1 \times 1$$

SOSTITUISCO E₃ nella (3)

$$16j + 4 + (-4j)E_2 + (12j + 4)\left(\frac{1}{4}j - 2 + E_2 - \frac{1}{4}jE_2\right) = 0$$

$$16j + 4 + (-4j)E_2 - 3 - 24j + 12jE_2 + 3E_2 + j - 8 + 4E_2 - jE_2 = 0$$

$$-4jE_2 - jE_2 + 12jE_2 + 3E_2 + 4E_2 = -16j - 4 + 3 + 24j - j + 8$$

$$E_2(-4j - j + 12j + 3 + 4) = 7j + 7$$

$$(7j + 7)E_2 = 7j + 7$$

$$E_2 = 1$$

$$I_{X1} = -1$$

quindi:

$$\begin{cases} E_1 = -2 \\ E_2 = 1 \\ E_3 = -1 \\ I_{X1} = -1 \end{cases}$$

BILANCIO DI POTENZA

$$V_{R1} = -E_3 = 1$$

$$V_{C1} = V_{R2} = E_1 - E_3 = -1$$

$$V_{C2} = E_2 - E_3 = 2$$

$$V_{R3} = E_2 = 1$$

$$V_{g1} = -2 \text{ (note)}$$

$$V_{g2} = E_2 - E_1 = 3$$

$$I_{R1} = \frac{V_{R1}}{R1} = \frac{1}{1} = 1$$

$$I_{C1} = \frac{V_{C1}}{\frac{1}{j\omega C1}} = \frac{-1}{\frac{1}{8j}} = -8j$$

$$I_{R2} = \frac{V_{R2}}{R2} = \frac{-1}{\frac{1}{2}} = -2$$

$$I_{C1} + I_{R2} = \frac{V_{C1}}{Z_{oc}} = \frac{-1}{\frac{1}{34} - \frac{2}{17}j} = \frac{34}{1-4j} \cdot \frac{1+4j}{1+4j} = \frac{34-136j}{17} = -2-8j$$

$$I_{C2} = \frac{V_{C2}}{\frac{1}{j\omega C2}} = \frac{2}{\frac{1}{4j}} = -8j$$

$$I_{R3} = \frac{V_{R3}}{R3} = \frac{1}{1} = 1$$

$$I_{g1} = I_{X1} = -1$$

$$I_{g2} = 1+8j \text{ (note)}$$

$$P_{CVg1} = \frac{1}{2} V_{g1} I_{g1}^* = \frac{1}{2} - 2 - 1 = -1$$

$$P_{Clg2} = \frac{1}{2} V_{g2} I_{g2}^* = \frac{1}{2} 3(4-8j) = \frac{3-24j}{2} = \frac{3}{2} - 12j$$

$$P_{AR1} = \frac{1}{2} R_1 |I_{R1}|^2 = \frac{1}{2} \cdot \frac{1}{2} \cdot 4 = 1$$

$$P_{AR2} = \frac{1}{2} R_2 |I_{R2}|^2 = \frac{1}{2} \cdot \frac{1}{2} \cdot 4 = 1$$

$$P_{AR3} = \frac{1}{2} R_3 |I_{R3}|^2 = \frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$$

$$\rightarrow 1 + \frac{3}{2} = 2 + \frac{1}{2}$$

$$\frac{5}{2} = \frac{5}{2}$$



$$Q_{C1} = -\frac{1}{2} \omega C_1 |V_{C1}|^2 = -\frac{1}{2} \cdot 2 \cdot 4 \cdot 1 = -4$$

$$Q_{C2} = -\frac{1}{2} \omega C_2 |V_{C2}|^2 = -\frac{1}{2} \cdot 2 \cdot 2 \cdot 4 = -8$$

$$\rightarrow -8 - 4 = -12$$

$$P_{Rg1} = \text{Im}(P_{CVg1}) = 0$$

$$P_{Rg2} = \text{Im}(P_{Clg2}) = -12$$



